

QUANTITATIVE FINANCE LEARNING PATH

The Complete Professional Guide

From Foundation to Interview Ready in 42 Weeks

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A comprehensive roadmap covering **26 professional resources** across Mathematics, Python, Regulatory, Models, Products, P&L Attribution, and Interview Prep.

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Your Journey at a Glance



 **Interview Ready: 26 Resources — 57 Products**
43 Projects — 250+ Problems

★ Why This Learning Path is Different

Unlike Free YouTube/Blogs:

- Desk-first, not textbook-first approach
- P&L attribution as core skill, not afterthought
- Mental math drills for interview composure
- Real interview questions from tier-1 firms
- Model failure patterns, not just derivations

Unlike Academic Courses:

- 43 implementation projects with code
- Practitioner insights from trading floors
- Focus on WHY models fail, not just HOW
- Regulatory knowledge (Basel, FRTB, XVA)
- Interview prep with timing drills

Quick Start Guide - Read This First

How to Use This Document

🎯 This Learning Path Is Designed For:

- Aspiring quants from mathematics, physics, engineering, computer science, or finance backgrounds
- Career switchers looking to transition into quantitative finance
- Students preparing for quant interviews at tier-1 firms
- Professionals seeking to deepen their understanding of derivatives pricing and risk management

🏆 What You'll Achieve:

- Master mathematical foundations: Linear Algebra, Differential Equations, Stochastic Calculus, Statistics, ML
- Understand pricing models for all major asset classes: IR, FX, Equity, Credit, Inflation
- Learn 57+ derivative products from vanillas to exotics including hybrids
- Master P&L attribution and desk diagnostics for real trading floor analysis
- Develop mental math and market intuition for rapid-fire interviews
- Build 43 industry-grade projects demonstrating practical implementation skills
- Prepare for interviews with 250+ solved problems from real quant interviews

How to Navigate This Guide

1. **Start with Phase 1** - Everyone begins here regardless of background
2. **Follow the phase sequence** - Each phase builds on previous knowledge
3. **Use coupon codes** - Apply the codes when purchasing resources for discounts
4. **Track your progress** - Use the success metrics at the end of each phase
5. **Adjust based on background** - See Section 8 for customized paths

Time Commitment & Timeline

Track	Details
Standard Track	42 weeks (10 months) at 20-25 hours/week
Accelerated Track	26-28 weeks (6.5 months) at 30-35 hours/week, focusing on 2-3 asset classes
Part-Time Track	56-60 weeks (13-15 months) at 10-15 hours/week
Intensive Track	18-22 weeks (5 months) for candidates with strong math background

+ What's New in This Edition

★ Four Critical New Resources Added

1. Python for Quants (Phase 2) — Code: PYTHON10

85+ pages covering Python as financial logic: speed, memory, GIL, `--slots--`, and system design with 50+ interview questions.

2. Regulatory & Risk Framework Primer (Phase 2) — Code: RISK10

45+ pages covering Basel I-IV, FRTB IMA/SA, CVA capital, VaR, ES, CCR, and 25+ interview questions.

3. P&L Attribution & Desk Diagnostics (Phase 3) — Code: PNL10

47 pages covering how to read P&L like a model engineer and diagnose model failures.

4. Mental Math & Market Intuition Drills (Phase 7) — Code: EXOTIC10

49 pages of rapid-fire techniques for option bounds, the 0.4 rule, and FX crosses.

\$ Investment Overview

All resources offer coupon codes (10% discount on each). Purchasing the complete bundle is more cost-effective than individual purchases. Each resource includes:

- 🕒 Lifetime access with no expiration
- 📄 Downloadable PDF format
- 🔄 All future updates and additions
- 📁 Desk-oriented, practical content

🛒 Recommended Purchase Strategy:

1. Buy Phase 1 resources immediately (Roadmap, Starter Pack, Probability)
2. Purchase Phase 2 foundations as you begin (LADE, Stats, Stochastic, ML, **Python**, **Regulatory**)
3. Add Desk Essentials in Week 11
4. Buy P&L Attribution & Desk Diagnostics in Week 12
5. Buy model and derivative notes as you reach each asset class
6. Get Projects Pack by Week 30
7. Purchase Mental Math & Interview Problems by Week 38

1 🧭 Phase 1: Foundation & Direction (Weeks 1–2)

Objective: Understand the quant landscape, build intuition, and assess your starting point.

What you'll learn:

- How quant roles differ across asset classes and desks
- What mathematical and technical skills are required

- How to map your current background to the learning path
- Core probability concepts used daily on trading desks
- Real-world analogies that make abstract concepts concrete

1.1 Resources (Purchase All Three)

Quant ROADMAP

📌 Foundational

📄 Code: CAREER10

Structured path for transitioning from any background into quantitative finance

Covers role definitions, skill requirements, timeline expectations, and career progression strategies.

[🔗 Click Here to Access](#)

Quant Finance Starter Pack

📌 Foundational

📄 Code: QUANT10

200+ pages of intuition-building content with real-world analogies

Covers desk culture, terminology, market mechanics, and foundational concepts without heavy mathematics.

[🔗 Click Here to Access](#)

Probability Quick Tricks

📌 Foundational

📄 Code: PROB10

Essential probability techniques and shortcuts used on trading desks

Includes combinatorics, conditional probability, expectation tricks, and common interview patterns.

[🔗 Click Here to Access](#)

1.2 Week-by-Week Breakdown

📅 Week 1:

- Days 1-2: Read Quant ROADMAP completely, identify your background profile
- Days 3-5: Study first 100 pages of Starter Pack, focus on market structure and terminology
- Days 6-7: Begin Probability Quick Tricks, work through basic examples

Week 2:

- Days 8-10: Complete Starter Pack, make notes on concepts that seem unclear
- Days 11-12: Finish Probability Quick Tricks, attempt all practice problems
- Days 13-14: Review and consolidate, create your personalized learning plan using Section 8

1.3 Success Metrics - Phase 1

✔ You're Ready for Phase 2 If You Can:

- Explain what quants do across different desks (structuring, trading, model validation)
- Describe the difference between vanilla and exotic derivatives
- Calculate conditional probabilities and expectations without looking up formulas
- Understand basic market terminology (bid/ask, vol surface, Greeks, curve)
- Articulate your learning goals and timeline based on your background

2 Phase 2: Core Mathematical Foundation + Python + Regulatory (Weeks 3–10)

Objective: Build the mathematical toolkit required for derivatives pricing and risk management.

What you'll learn:

- Linear algebra for PCA, dimensionality reduction, and factor models
- Differential equations for PDEs in option pricing
- Statistical inference, hypothesis testing, and model risk quantification
- Stochastic calculus including Itô's lemma, martingales, and change of measure
- Machine learning applications in trading, risk management, and alpha generation
- **NEW:** Python for quants - speed, memory, GIL, system design, 50+ interview questions
- **NEW:** Regulatory frameworks - Basel I-IV, FRTB, CVA capital, VaR, Expected Shortfall

Why this phase matters: You cannot understand why models fail without understanding the mathematics behind them. Desk quants need to know when to trust a model and when to override it.

2.1 Resources (Purchase All Six)

Linear Algebra & Differential Equations for Quants

📌 Mathematics

📖 Code: LADE10

60+ pages covering PCA, eigenvalue problems, ODEs, and PDEs

Desk-oriented with practical applications including rates curves, correlation matrices, and bond pricing.

[🔗 Click Here to Access](#)

Statistics & Econometrics for Quants

📌 Mathematics

📖 Code: STATS10

Comprehensive desk-first statistical inference and model risk assessment

Covers backtesting frameworks, hypothesis testing in trading strategies, and econometric applications.

[🔗 Click Here to Access](#)

Stochastic Calculus for Quants

📌 Mathematics

📖 Code: STOCHASTIC10

Itô calculus explained in desk language

Covers Brownian motion, martingales, Girsanov theorem, Feynman-Kac, and derivatives pricing applications.

[🔗 Click Here to Access](#)

Machine Learning for Quants

📌 Mathematics

📖 Code: ML10

Desk-first ML for trading signals, risk models, and portfolio optimization

Includes supervised/unsupervised learning, volatility forecasting, and 30+ interview questions.

[🔗 Click Here to Access](#)

Python for Quants

🔖 Programming - NEW

||||| Code: PYTHON10

Complete interview guide covering speed, memory, and system design

85+ pages covering Python as financial logic: data structures, NumPy vectorization, GIL and concurrency, memory optimization with `__slots__`, and system design patterns. 50+ interview questions with model answers.

[Click Here to Access](#)

Regulatory & Risk Framework Primer

🔖 Risk Management - NEW

||||| Code: RISK10

Master Basel to FRTB with desk-first regulatory intuition

45+ pages covering Basel I-IV evolution, FRTB IMA/SA, CVA capital charges, market risk metrics (VaR, ES, Stress VaR), CCR, operational risk, and 25+ interview questions. Real interview scenarios and memory anchors included.

[Click Here to Access](#)

2.2 Week-by-Week Breakdown

Weeks 3-4: Linear Algebra & Differential Equations

- Study matrix operations, eigenvalues/eigenvectors, SVD decomposition
- Learn PCA application to yield curve construction
- Master first-order and second-order ODEs
- Understand Black-Scholes PDE derivation
- Practice problems: 20-30 from interview questions pack

Weeks 5-6: Statistics & Econometrics

- Master probability distributions (normal, lognormal, Student-t, chi-squared)
- Learn maximum likelihood estimation and method of moments
- Study hypothesis testing, confidence intervals, p-values
- Understand time series analysis: AR, MA, ARMA, GARCH
- Learn regression analysis and model diagnostics
- Practice problems: 30-40 from interview questions pack

Weeks 7-8: Stochastic Calculus

- Master Brownian motion properties and simulation
- Learn Itô's lemma and its applications
- Understand martingales and stopping times
- Study Girsanov theorem and change of measure
- Learn Feynman-Kac formula for PDE solutions
- Practice problems: 25-35 from interview questions pack

Weeks 9-10: Machine Learning

- Study supervised learning: regression, classification, tree-based methods
- Learn unsupervised learning: clustering, dimensionality reduction
- Understand cross-validation, overfitting, regularization
- Study neural networks and deep learning basics
- Learn ML applications in vol surface construction and alpha generation
- Practice the 30+ ML interview questions included in the notes

2.3 Success Metrics - Phase 2

✔ You're Ready for Phase 3 If You Can:

- Derive Itô's lemma and apply it to derive the Black-Scholes PDE
- Explain why martingales are central to risk-neutral pricing
- Perform PCA on a yield curve and interpret the first 3 components
- Design a hypothesis test for backtesting a trading strategy
- Implement basic ML models (linear regression, random forest) from scratch
- Solve 80%+ of mathematical problems in the interview questions pack without hints

3 Phase 3: Desk Essentials & P&L Mastery (Weeks 11–15)

Objective: Understand how theoretical knowledge translates to actual desk workflows, risk management, P&L attribution, and model diagnostics.

What you'll learn:

- Greeks (Delta, Gamma, Vega, Theta, Rho) and their practical implications
- Volatility surfaces: construction, interpolation, and arbitrage-free conditions
- Yield curve construction and bootstrapping techniques
- Time series analysis for trading signals and risk factors

- Monte Carlo simulation for pricing and risk management
- XVA (CVA, DVA, FVA, KVA) and counterparty risk
- **NEW:** P&L as model validation, not arithmetic
- **NEW:** Clean vs Dirty P&L and theta bleed diagnostics
- **NEW:** Greeks that lie and correlation breakdown patterns
- **NEW:** Residual P&L analysis and model reserves

⚠ Critical Phase

This is where you transition from **academic quant** to **desk quant**. Understanding P&L attribution and model diagnostics separates those who can price from those who can trade.

3.1 Resources (Purchase Both)

Desk Essentials

🔑 Desk Skills

📄 Code: DESK10

Comprehensive guide for real trading desk workflows

Greeks computation, volatility surfaces, yield curves, Monte Carlo methods, and XVA framework with practical examples.

[🔗 Click Here to Access](#)

P&L Attribution & Desk Diagnostics

🔑 Desk Skills - NEW

📄 Code: PNL10

47-page guide to reading P&L like a model engineer

Clean vs dirty P&L, theta bleed, Greeks that lie, correlation breakdown, residual analysis, and forensic reports.

[🔗 Click Here to Access](#)

3.2 Week-by-Week Breakdown

📅 Week 11: Greeks & Volatility

- Master analytical Greeks computation for vanilla options
- Learn numerical Greeks calculation (bump-and-revalue, finite differences)
- Study volatility smile and term structure
- Understand implied volatility calculation and its challenges
- Learn practical hedging strategies using Greeks

Week 12: Curves & Time Series

- Master yield curve bootstrapping from swaps, bonds, and futures
- Learn interpolation methods (linear, cubic spline, Nelson-Siegel)
- Study forward curve construction and no-arbitrage conditions
- Understand time series models for rates, FX, and volatility
- Learn how to calibrate ARMA and GARCH models

Week 13: Monte Carlo & XVA

- Master Monte Carlo path generation for various processes
- Learn variance reduction techniques (antithetic variates, control variates)
- Study importance sampling and quasi-random sequences
- Understand CVA calculation and wrong-way risk
- Learn FVA, KVA concepts and their impact on pricing

Week 14: P&L Attribution Fundamentals

- **Module 1-3:** P&L as model validation, clean vs dirty P&L split, theta bleed diagnostics
- **Module 4-6:** Why Greeks lie (barriers, vol spikes), hedge slippage, correlation breakdown
- Master the diagnostic mindset: residual P&L is sacred, Greeks are suspect
- Practice: Analyze 5 sample P&L reports from the guide

Week 15: Advanced P&L Diagnostics

- **Module 7-10:** Residual P&L patterns, trader disputes, stress P&L, model reserves
- **Module 11-13:** Case studies (barrier bleed, CMS shock, Bermudan mis-exercise)
- Learn the 5-minute triage framework for unexplained P&L
- Practice: Diagnose 10 P&L mysteries from the forensic reports section

3.3 Success Metrics - Phase 3

✔ You're Ready for Phase 4 If You Can:

- Calculate and interpret all Greeks for a portfolio of vanilla options
- Bootstrap a yield curve from market instruments and explain interpolation choices
- Construct an arbitrage-free volatility surface from market quotes
- Implement Monte Carlo pricing with variance reduction for path-dependent options
- Explain how XVA adjustments affect derivatives pricing and hedging
- **NEW:** Separate clean vs dirty P&L and explain what each component reveals
- **NEW:** Diagnose whether P&L loss is execution, liquidity, or model slippage
- **NEW:** Spot when Greeks are lying (near barriers, high vol-of-vol, illiquid markets)
- **NEW:** Identify correlation concentration before it explodes
- **NEW:** Explain why residual P&L $\geq 5\%$ is a model failure
- Describe how a desk would hedge a structured product end-to-end

🏆 Critical Checkpoint

After Phase 3, you understand **HOW** desks operate day-to-day, **WHY** mathematical models matter in practice, and **WHEN** models fail. You can read P&L like a model engineer and diagnose issues faster than traders. You're ready to learn specific pricing models for each asset class.

4 🧩 Phase 4: Asset Class Models (Weeks 16–22)

Objective: Master pricing models for all major asset classes, understanding when they work and when they fail.

Critical Note: Study models in the order listed. Interest rates is foundational for all other asset classes. Each model note explains calibration, implementation, limitations, and model risk. With your P&L attribution knowledge, you'll now understand how model choices impact daily P&L.

4.1 Why This Order?

- **Interest Rates First:** Discount curves and rates dynamics underpin all derivatives pricing
- **FX Second:** Builds on rates models, introduces quanto adjustments and cross-currency concepts
- **Equity Third:** More standalone, but dividend forwards require rates knowledge
- **Credit Fourth:** Combines rates (discounting) with equity-like dynamics (default intensity)

4.2 Resources (Purchase All Four)

Interest Rates Models for Quants

Models

 Code: RATES10

Deep dive into Hull-White, LMM, SABR, and short-rate models

Calibration procedures, model failure patterns, and model risk management on rates desks.

[Click Here to Access](#)

FX Models for Quants

Models

 Code: FXD10

Garman-Kohlhagen, local vol, stochastic vol, and jump-diffusion for FX

Carry dynamics, FX smile, quanto adjustments, cross-currency considerations, and FX-specific model risk.

[Click Here to Access](#)

Equity Models for Quants

Models

 Code: EQUITIES10

From Black-Scholes to local vol, stochastic vol (Heston), and jump-diffusion

Crash modeling, skew dynamics, vol-of-vol, dividend modeling, and equity-specific model risk.

[Click Here to Access](#)

Credit Models for Quants

Models

 Code: CREDITS10

Intensity-based, structural, and reduced-form credit models

CDS pricing, default correlation, jump-to-default, tranche valuation, and credit-specific model risk.

[Click Here to Access](#)

4.3 Week-by-Week Breakdown

Weeks 16-17: Interest Rates Models

- Master Hull-White 1-factor: calibration to caps/floors, implementation, mean reversion
- Learn Hull-White 2-factor for decorrelated term structure moves
- Study LIBOR Market Model (LMM) for exotic rates derivatives
- Understand SABR model for capturing swaption volatility smile
- Learn when each model is appropriate and limitations
- **Connect to P&L:** How Hull-White parameters drift affects dirty P&L
- Implement basic versions in Python/C++

Weeks 18-19: FX Models

- Master Garman-Kohlhagen model for vanilla FX options
- Learn FX smile modeling: risk reversals and butterflies
- Study stochastic volatility models for FX
- Understand quanto adjustments and cross-currency dynamics
- Learn jump-diffusion models for FX crises
- Study FX carry strategies and their risks
- **Connect to P&L:** Sticky-strike vs sticky-delta and delta hedge slippage

Weeks 20-21: Equity Models

- Understand Black-Scholes assumptions and where they break
- Master local volatility models (Dupire equation)
- Learn Heston model for stochastic volatility
- Study jump-diffusion (Merton, Kou) for crash modeling
- Understand equity skew dynamics and term structure
- Learn dividend modeling and forward pricing
- **Connect to P&L:** How vanna/volga corrections prevent delta slippage

Week 22: Credit Models

- Master intensity-based (reduced-form) models
- Learn structural models (Merton, Black-Cox)
- Study CDS pricing and curve bootstrapping
- Understand default correlation and copula models
- Learn jump-to-default modeling
- Study CDO tranche pricing
- **Connect to P&L:** Jump-to-default creates residual P&L spikes

4.4 Success Metrics - Phase 4

✔ You're Ready for Phase 5 If You Can:

- Calibrate Hull-White model to market cap/floor prices
- Explain why SABR is better than Black-Scholes for swaptions
- Price an FX option using Garman-Kohlhagen and adjust for quanto if needed
- Implement Heston model and explain vol-of-vol impact
- Bootstrap a CDS curve and price a first-to-default basket
- Articulate when each model is appropriate and what risks it ignores
- Explain model risk for each asset class with concrete examples
- **NEW:** Predict how model parameter drift will appear in P&L attribution
- **NEW:** Identify which models create persistent residual bias

5 Phase 5: Derivatives Products (Weeks 23–30)

Objective: Learn the complete product lifecycle from payoff definition to model selection to P&L attribution for 57 derivative products across all asset classes.

Critical Note: Study products AFTER models for each asset class. This ensures you understand WHY certain models are chosen for specific products. Your P&L attribution knowledge will help you understand how each product's Greeks behave in practice.

5.1 Product → Model → P&L Mapping

Each derivative note follows this structure:

1. **Product Definition:** Payoff structure, market conventions, typical use cases
2. **Model Selection:** Which model(s) to use and why, calibration approach
3. **Pricing & Hedging:** Greeks calculation, hedging strategy, model risk
4. **P&L Attribution:** How to explain daily P&L to risk management
5. **Market Reality:** Bid/ask spreads, liquidity considerations, desk perspective

5.2 Resources (Purchase All Five)

Interest Rate Derivatives

Products

 Code: IR10

15 products from IRS to exotic structures

Complete mapping from product to model to hedging strategy for rates desks.

[Click Here to Access](#)

FX Derivatives

Products

 Code: FX10

12 products including barriers, digitals, and PRDC

Cross-currency and emerging market considerations with practical examples.

[Click Here to Access](#)

Equity Derivatives

Products

 Code: EQUITY10

12 products from vanillas to autocalls and variance swaps

Structured products and exotics explained with skew and correlation dynamics.

[Click Here to Access](#)

Credit Derivatives

Products

 Code: CREDIT10

10 products including CDS, CDO, and baskets

Default correlation, tranche valuation, and jump-to-default modeling.

[Click Here to Access](#)

Inflation Derivatives

Products

 Code: INFLATION10

8 products covering ZCIS, YoY swaps, and TIPS

Seasonality, indexation lags, and real vs nominal rate dynamics.

[Click Here to Access](#)

5.3 Week-by-Week Breakdown

Weeks 23-24: IR Derivatives (15 products)

- Week 23: IRS, FRAs, caps/floors, swaptions, CMS (8 products)
- Week 24: Callable bonds, range accruals, Bermudan swaptions, PRDC, exotics (7 products)
- Focus: Curve construction, convexity adjustments, path-dependency
- **P&L Connection:** Why Bermudan swaptions create model theta vs true theta confusion

Weeks 25-26: FX Derivatives (12 products)

- Week 25: Spot, forwards, NDFs, vanilla options, barriers (6 products)
- Week 26: Digitals, target forwards, PRDC, quanto, exotics (6 products)
- Focus: Smile dynamics, cross-currency basis, quanto adjustments
- **P&L Connection:** How FX barriers create gamma explosions that Greeks can't predict

Weeks 27-28: Equity Derivatives (12 products)

- Week 27: Vanilla options, barriers, autocalls, cliquets (6 products)
- Week 28: Variance swaps, volatility swaps, worst-of, dispersion, exotics (6 products)
- Focus: Skew, vol-of-vol, correlation, dividend risk
- **P&L Connection:** Why worst-of baskets suffer correlation breakdown in stress

Week 29: Credit Derivatives (10 products)

- CDS (single-name), CDS indices, index tranches (5 products)
- First-to-default, nth-to-default, options on credit, TRS, CDOs (5 products)
- Focus: Default correlation, recovery rates, jump-to-default risk
- **P&L Connection:** Jump-to-default creates residual spikes that can't be hedged

Week 30: Inflation Derivatives (8 products)

- ZCIS, YoY swaps, inflation caps/floors, TIPS (4 products)
- Inflation-linked bonds, exotic structures, real rate dynamics (4 products)
- Focus: Seasonality, indexation lags, real vs nominal rates
- **P&L Connection:** How CPI seasonality creates theta patterns that confuse traders

5.4 Success Metrics - Phase 5

✔ You're Ready for Phase 6 If You Can:

- Explain the payoff structure of any of the 57 products to a non-quant
- Choose the appropriate pricing model for each product and justify your choice
- Calculate Greeks and design a hedging strategy for vanilla and exotic products
- Explain P&L attribution for a portfolio containing multiple products
- Identify model risk for each product and suggest model improvements
- Describe how market conditions affect product pricing and hedging
- Answer product-related interview questions with confidence
- **NEW:** Predict which products will have large residual P&L and why
- **NEW:** Explain how each product's Greeks lie in different market regimes

6 </> Phase 6: Practical Implementation (Weeks 31–38)

Objective: Build 43 industry-grade projects that demonstrate your ability to implement pricing models, risk systems, and trading strategies.

Why Projects Matter: Interviews increasingly include take-home assignments and coding assessments. A strong GitHub portfolio with production-quality code differentiates you from other candidates.

6.1 Resource

Quant Projects Pack

📁 Implementation

📄 Code: PROJECT10

43 industry-grade projects across three difficulty levels

10 Foundational, 10 Moderate, 10 Advanced, plus 13 asset-class-specific implementations with full code and documentation.

[🔗 Click Here to Access](#)

6.2 Project Structure

Foundational Level (10 projects):

- Project 1: One-Period Binomial Model and Arbitrage Detection
- Project 2: GBM Simulation and Monte-Carlo Pricing of European Options
- Project 3: Historical and EWMA Volatility Estimation
- Project 4: American Put Pricing via Binomial Model
- Project 5: Barrier Hitting Probability Simulation
- Project 6: Asian Option Pricing via Monte-Carlo
- Project 7: Euler-Maruyama Simulation of SDEs
- Project 8: Implied Volatility Computation Using Newton's Method
- Project 9: Exchange Option Pricing via Monte-Carlo
- Project 10: Numerical Greeks via Finite Differences

Moderate Level - Core Pricing Methods (10 projects):

- Project 1: Multi-Step Binomial Tree for European, American, and Exotic Options
- Project 2: Trinomial Tree with Adaptive Branching and Barrier Support
- Project 3: Crank-Nicolson Finite-Difference PDE Solver
- Project 4: Robust Finite-Difference Greeks with Richardson Extrapolation
- Project 5: Merton Jump-Diffusion Monte-Carlo with Variance Reduction
- Project 6: Heston Stochastic Volatility Monte-Carlo with Exact Simulation
- Project 7: Local Volatility Surface via Dupire Formula
- Project 8: Implied Volatility Surface Construction
- Project 9: American Barrier Option Pricing via Monte-Carlo Regression
- Project 10: Calibration of Heston Model to Market IV Surface

Advanced Level (10 projects):

- Project 1: Hull-White One-Factor Model Calibration
- Project 2: HJM Multi-Factor Monte-Carlo Simulation Engine
- Project 3: FX Vanilla Volatility Smile Adjustments and AD
- Project 4: Reduced-Form Credit Risk via Jarrow-Turnbull
- Project 5: Regulatory Credit Portfolio VaR via Gaussian & t-Copulas
- Project 6: CVA/DVA via Monte-Carlo Exposure Simulation
- Project 7: Local-Stochastic Volatility (LSV) Model
- Project 8: Regime-Switching Volatility Model with Hamilton Filtering
- Project 9: Commodity Derivatives with Seasonal Stochastic Models
- Project 10: Advanced XVA Framework - FVA, MVA, KVA

Asset-Class Specific (13 projects):

Interest Rate Projects (3): Zero-Coupon Curve Construction, Bermudan Swaption Pricing, Quanto IR Swap

FX Projects (3): FX Forward Curve Bootstrapping, Exotic FX Option, Quanto Equity Option

Equity Projects (3): Dividend-Adjusted Black-Scholes, American Asian Option, Equity-Credit Hybrid

Bond Projects (3): YTM/Duration/Convexity Engine, Callable Bond Pricing, Inflation-Linked Bond

Inflation Projects (1): CPI Simulation & Zero-Coupon Inflation Swap Pricing

6.3 Week-by-Week Implementation Plan

Weeks 31-32: Foundational Projects (10 projects)

- Complete 1 project per day (5 per week)
- Focus on code quality: documentation, testing, modularity
- Create GitHub repository with professional README
- Each project should take 3-5 hours
- **P&L Integration:** Add P&L attribution module to each pricer

Weeks 33-35: Moderate + Asset-Specific IR & FX (16 projects)

- Week 33: Moderate projects 1-5 (5 projects)
- Week 34: Moderate projects 6-10 (5 projects)
- Week 35: IR projects (3) + FX projects (3)
- Each project should take 5-8 hours
- **P&L Integration:** Implement clean vs dirty P&L separation

Weeks 36-37: Asset-Specific Equity, Bond, Inflation (7 projects)

- Week 36: Equity projects (3) + Bond projects (3)
- Week 37: Inflation project (1) + integration testing (6 days)
- Each project should take 6-8 hours
- **P&L Integration:** Add residual P&L analysis and diagnostics

Week 38: Advanced Projects (10 projects)

- Complete 1-2 advanced projects per day
- Focus on optimization, performance, real-time capabilities
- Integrate multiple components into unified frameworks
- Each project should take 8-12 hours
- **P&L Integration:** Build comprehensive P&L attribution system

6.4 Best Practices for Project Implementation

1. **Version Control:** Use Git from day one, commit frequently with clear messages
2. **Code Structure:** Separate data, models, analytics, and visualization layers
3. **Documentation:** Write clear README files with usage examples and mathematical background
4. **Testing:** Include unit tests comparing your results to analytical solutions
5. **Performance:** Profile your code, optimize bottlenecks, use vectorization
6. **Visualization:** Create clear plots for Greeks, P&L, convergence analysis
7. **Error Handling:** Implement robust error checking and edge case handling
8. **Extensibility:** Design code that can be easily extended for variations
9. **NEW: P&L Attribution:** Every pricer should output clean P&L, dirty P&L, and residual

6.5 Programming Languages & Tools

Primary Languages:

- **</>** Python: NumPy, SciPy, Pandas, Matplotlib
- **</>** C++: For performance-critical components
- **</>** R: Statistical analysis and econometrics

Essential Libraries:

- QuantLib for derivatives pricing
- TensorFlow/PyTorch for ML
- Jupyter notebooks for analysis
- Git/GitHub for portfolio

6.6 Success Metrics - Phase 6

✔ You're Ready for Phase 7 If You:

- Have completed 40+ projects with clean, documented code
- Maintain a GitHub portfolio that showcases your work
- Can explain design decisions and trade-offs in your implementations
- Have tested your code against analytical solutions and benchmark results
- Can optimize code for performance (runtime, memory usage)
- Understand when to use Python vs C++ for different problems
- Can walk through your code in a technical interview setting
- Have integrated multiple components (pricing + risk + P&L)
- **NEW:** Every project outputs clean vs dirty P&L separation
- **NEW:** Can demonstrate residual P&L pattern detection in your code

7 Phase 7: Interview Mastery (Weeks 39–42)

Objective: Master mental math techniques, rapid-fire problem solving, and 250+ interview questions covering mathematics, products, models, coding, and behavioral topics.

What's New: This phase now integrates intensive mental math training to develop composure under pressure. You'll learn to bound problems instantly, use approximation frameworks, and never freeze during interviews.

! The Interview Reality

Interviewers don't want exact answers—they want to see how you think under pressure. The candidate who says "Let me bound this first: max is spot, min is zero" beats the one who silently tries to compute Black-Scholes and freezes.

7.1 Resources (Purchase Both)

Mental Math & Market Intuition Drills

 Interview Prep - NEW

 Code: EXOTIC10

49-page guide with 50+ rapid-fire drills

Option pricing bounds, implied vol checks, the 0.4 rule, FX crosses, WRAPS framework, and anti-freeze tactics.

[Click Here to Access](#)

250+ Quant Interview Problems

 Interview Prep

 Code: FIRST10

Comprehensive collection with solved examples and practice sheets

Covers probability, stochastic calculus, derivatives pricing, brainteasers, coding challenges, and behavioral questions.

[Click Here to Access](#)

7.2 Interview Question Categories

Category	Questions
Mathematical Foundations	70 questions
• Probability and combinatorics	25 questions
• Stochastic calculus and martingales	20 questions
• Linear algebra and ODEs/PDEs	15 questions
• Statistics and time series	10 questions
Derivatives Pricing	60 questions
• Black-Scholes and Greeks	20 questions
• Interest rate products	15 questions
• Exotic options and barriers	15 questions
• Credit and correlation products	10 questions
Coding & Implementation	50 questions
• Monte Carlo simulation	15 questions
• Numerical methods and optimization	15 questions
• Data structures and algorithms	10 questions
• System design for trading systems	10 questions
Brain Teasers & Puzzles	40 questions
Behavioral & Fit	30 questions

7.3 Week-by-Week Interview Prep

📅 Week 39: Mental Math Mastery

Day 1-2: Master option pricing bounds - the safety rails (Module 1)

- Memorize: $0 \leq C \leq S$, $0 \leq P \leq K$, Put-Call Parity
- Practice 20 bound-checking drills with 30-second timer
- Learn to spot arbitrage in 5 seconds

Day 3-4: Implied vol sanity checks and ranges (Module 2)

- Memorize vol ranges: EURUSD 6-12% (normal), SPY 12-20% (normal)
- Learn smile structure rules (RR, BF must be positive)
- Practice 15 vol regime identification drills

Day 5-6: The 0.4 rule and quick approximations (Module 3)

- Master: ATM Call $\approx 0.4 \times \sigma \times S \times \sqrt{T}$
- Memorize $\sqrt{T}$ table: 3M=0.5, 6M=0.71, 1Y=1.0
- Practice 25 pricing drills with moneyness adjustments

Day 7: FX crosses, yield curves, and 30-second drill library (Modules 4-7)

- Master triangular arbitrage detection
- Learn forward rate extraction shortcuts
- Complete 30 mixed rapid-fire drills

📅 Week 40: Mathematical Foundations & Products

- Day 1-2: Probability problems (25 questions)
- Day 3-4: Stochastic calculus problems (20 questions)
- Day 5-6: Product pricing and Greeks (35 questions)
- Day 7: Integration practice - combine mental math with rigorous solutions
- **Key Practice:** For each problem, first bound it (5 sec), then solve rigorously
- Target: Solve 60% without hints, 90% with hints

📅 Week 41: Coding & Implementation

- Day 1-2: Monte Carlo problems (15 questions)
- Day 3-4: Numerical methods (15 questions)
- Day 5-6: Algorithms and data structures (10 questions)
- Day 7: System design (10 questions) + P&L attribution systems
- Target: Code solutions cleanly with proper testing
- **NEW:** Every coding solution should include P&L diagnostics output

Week 42: Full Mock Interviews & Polish

- Day 1: Brain teasers (40 questions) - timed practice with mental math techniques
- Day 2: Behavioral preparation and story development
- Day 3-4: Full mock interview simulations (2 hours each)
 - Practice the anti-freeze protocol: "Let me bound this first..."
 - Use WRAPS framework for trade quality questions
 - Demonstrate P&L diagnostic thinking in product questions
- Day 5-6: Review mistakes, identify weak areas
- Day 7: Final review and confidence building
 - Review the 5 key formulas (Module 9.1 of Mental Math)
 - Practice power phrases for freeze recovery
 - Visualize success scenarios

7.4 The Mental Math Framework - Critical Techniques

The Three Mental Math Principles:

1. **Principle 1:** Bound first, refine later
2. **Principle 2:** Use approximations the interviewer expects
3. **Principle 3:** Narrate your logic out loud

The 30-Second Answer Template:

1. Step 1 (5 sec): State the bounds
2. Step 2 (10 sec): Use approximation rule (0.4 rule, forward formula, etc.)
3. Step 3 (10 sec): Sanity check
4. Step 4 (5 sec): State confidence interval

The Freeze Recovery Protocol (when stuck):

- Second 1-2: "Let me bound this first..."
- Second 3-4: State max and min values
- Second 5: "So the answer is between X and Y"

Power Phrases to Memorize:

- "Let me bound this first..."
- "Rough answer: X, let me refine..."
- "I don't know the exact formula, but here's my approach..."
- "This can't be more than... and can't be less than..."
- "The intuition is... so I'd estimate..."
- "Let me sanity-check that..."

7.5 Interview Preparation Strategy

Technical Preparation:

1. Solve problems without looking at solutions first
2. If stuck after 20 minutes, check hints but not full solutions
3. After solving, review the official solution for better approaches
4. Re-solve difficult problems 2-3 days later
5. Practice explaining solutions out loud as if in an interview
6. **NEW:** Always start with mental math bounds before rigorous solution

Coding Interviews:

1. Use whiteboard or shared document, not IDE
2. Talk through your approach before coding
3. Write clean, modular code with clear variable names
4. Test your code with edge cases
5. Discuss time/space complexity
6. **NEW:** Always include P&L attribution in pricing code

Behavioral Preparation:

1. Prepare 3-5 project stories with clear technical depth
2. Know your resume inside-out; be ready to discuss any detail
3. Research the firm's trading strategies and recent news
4. Prepare thoughtful questions for interviewers
5. Practice the "why quant finance" story until it's natural
6. **NEW:** Prepare stories about diagnosing P&L mysteries in your projects

7.6 Common Interview Formats

Phone Screen (30-45 min):

- 1-2 probability/stochastic calculus questions
- 1 derivatives pricing question (use mental math first!)
- Basic behavioral questions

Technical Round (60-90 min):

- 2-3 math/probability questions
- 1-2 derivatives pricing questions
- Discussion of your projects (highlight P&L attribution features)

Coding Round (60-90 min):

- Implement a pricer or risk metric
- Monte Carlo or numerical methods question
- Algorithm/data structure problem

On-Site (4-6 hours):

- Multiple technical rounds with different team members
- Coding session with live implementation
- Case study or take-home problem discussion
- Behavioral/fit interview with hiring manager

7.7 The Interview Day Checklist

Night Before:

- Sleep 8 hours (mental math requires sharp thinking)
- Review vol ranges table and $\sqrt{T}$ table
- Write down power phrases on index card

Morning Of:

- Solve 3 option bounds problems (warm up)
- Recite: "Bound first, compute never (if possible)"
- Eat protein (avoid sugar crash)

30 Minutes Before:

- Deep breathing (4-7-8 technique)
- Recite the 5 key formulas out loud
- Visualize yourself bounding a question successfully
- Turn off phone

7.8 Success Metrics - Phase 7

You're Interview-Ready If You Can:

- **NEW:** Bound any option price in 10 seconds using max/min principles
- **NEW:** Identify vol regime (normal vs crisis) in 5 seconds from quotes
- **NEW:** Use 0.4 rule to price ATM options within 10% in 5 seconds
- **NEW:** Calculate FX crosses and detect triangular arbitrage in 15 seconds
- **NEW:** Never freeze - always use "Let me bound this first" protocol
- **NEW:** Apply WRAPS framework to any trade quality question in 30 seconds
- Solve 80%+ of mathematical problems within time limits
- Explain pricing approaches for all major derivative products
- Code clean implementations of pricers and risk metrics under pressure
- Articulate model limitations and when models fail
- Walk through your projects with technical depth, highlighting P&L diagnostics
- Answer behavioral questions naturally with specific examples
- Handle unexpected questions by reasoning through first principles
- Ask intelligent questions that show market awareness
- Demonstrate composure under pressure (mental math training pays off here)

You Are Ready

With mental math fluency, P&L diagnostic skills, and 250+ solved problems, you can handle any interview scenario. Remember: The interviewer wants to see how you think under pressure, not perfection. **Frameworks beat formulas. Process beats precision. You've got this.**

8 Customized Learning Paths by Background

This section provides tailored guidance based on your starting background. Follow your path while referring to the main phases for detailed content.

8.1 Path A: Strong Math, No Finance (e.g., Math/Physics PhD)

Your Strengths: Advanced mathematics, rigorous thinking, research skills
Your Gaps: Market mechanics, product knowledge, desk culture, P&L intuition
Timeline: 5-6 months (accelerated)

Recommended Sequence:

1. Phase 1: Roadmap + Starter Pack + Probability (Weeks 1-2)
2. **SKIP:** Most of Phase 2 math (you already know this)
3. Phase 2 (Modified): Stochastic Calculus + ML only (Weeks 3-4)
4. Phase 3: Desk Essentials + **P&L Attribution (CRITICAL)** (Weeks 5-8)
5. Phase 4: All Models (Weeks 9-14)
6. Phase 5: All Derivatives (Weeks 15-21)
7. Phase 6: Projects with heavy focus (Weeks 22-27)
8. Phase 7: Mental Math + Interview Prep (Weeks 28-31)

! Key Focus Areas:

- Spend extra time on Starter Pack to understand market intuition
- **CRITICAL:** P&L Attribution is your bridge from theory to practice
- Focus on model limitations and when they fail (not just derivations)
- Learn product conventions and market terminology
- Practice explaining concepts simply without jargon
- Mental math training is essential - your math skills won't help if you freeze

8.2 Path B: Finance Background, Weaker Math (e.g., MBA, Economics)

Your Strengths: Market intuition, product knowledge, business context
Your Gaps: Rigorous mathematics, programming, model implementation
Timeline: 10-12 months (extended)

Recommended Sequence:

1. Phase 1: All three resources (Weeks 1-2)
2. Phase 2: **DEEP DIVE** - all four resources with extra practice (Weeks 3-12)
3. Phase 3: Desk Essentials + P&L Attribution (Weeks 13-17)
4. Phase 4: All Models (Weeks 18-24)
5. Phase 5: All Derivatives (Weeks 25-32)
6. Phase 6: Projects (Weeks 33-42)
7. Phase 7: Mental Math + Interview Prep (Weeks 43-46)

! Key Focus Areas:

- Invest heavily in Phase 2 - this is your bottleneck
- Work through 50+ practice problems per week in Weeks 3-12
- Learn programming fundamentals (Python) alongside mathematics
- Leverage your market knowledge when studying products
- Focus on practical applications rather than abstract theory
- P&L Attribution will feel natural given your finance background - excel here
- Mental math will leverage your market intuition - this is your strength

8.3 Path C: CS/Engineering, Limited Finance (e.g., Software Engineer)

Your Strengths: Programming, algorithms, system design, ML
Your Gaps: Finance domain knowledge, stochastic calculus, derivatives
Timeline: 7-8 months

Recommended Sequence:

1. Phase 1: All three resources (Weeks 1-2)
2. Phase 2: LADE + Stats + Stochastic (Weeks 3-8, skip ML)
3. Phase 3: Desk Essentials + P&L Attribution (Weeks 9-13)
4. Phase 4: All Models (Weeks 14-19)
5. Phase 5: All Derivatives (Weeks 20-26)
6. Phase 6: **HEAVY FOCUS** on Projects (Weeks 27-34)
7. Phase 7: Mental Math + Interview Prep (Weeks 35-38)

! Key Focus Areas:

- Leverage coding skills in Phase 6 - build portfolio-worthy projects
- Focus on system design for trading and risk systems
- Learn finance-specific algorithms (tree methods, Monte Carlo)
- Understand business context of technical decisions
- Practice explaining technical concepts to non-technical audiences
- **STRENGTH:** Build best-in-class P&L attribution systems in your projects
- Mental math will require discipline - you can't code your way through interviews

8.4 Path D: Career Switcher, Limited Time (e.g., Working Professional)

Your Constraints: 10-15 hours per week, need results in 6-12 months
Your Strategy: Focus deeply on 1-2 asset classes, build expertise
Timeline: 12-15 months

Recommended Sequence:

1. Phase 1: Roadmap + Starter Pack (Weeks 1-2)

2. Phase 2: LADE + Stats + Stochastic only (Weeks 3-10)
3. Phase 3: Desk Essentials + P&L Attribution (Weeks 11-15)
4. **CHOOSE 1-2 ASSET CLASSES** (e.g., Equity + Credit)
5. Phase 4: Selected Models only (Weeks 16-20)
6. Phase 5: Selected Derivatives only (Weeks 21-26)
7. Phase 6: Projects for selected asset classes (Weeks 27-36)
8. Phase 7: Mental Math + Interview Prep (Weeks 37-40)
9. **EXPAND:** Add another asset class (Weeks 41-52)

! Key Focus Areas:

- Choose asset classes aligned with your interests and target roles
- Go very deep rather than broad - become expert in chosen areas
- Build 15-20 high-quality projects rather than 43 average ones
- Target roles specifically in your chosen asset classes
- Expand knowledge after landing first role
- P&L Attribution is non-negotiable even for focused path
- Mental math training fits well into commute time - practice daily

9 Daily & Weekly Study Recommendations

9.1 Daily Structure for Full-Time Study (20-25 hrs/week)

Morning Block (3 hours):

- **08:00-09:30:** Theory study (notes, derivations, concepts)
- **09:30-09:45:** Break and review
- **09:45-11:00:** Practice problems or worked examples

Afternoon Block (2.5 hours):

- **13:00-14:30:** Implementation (coding projects or exercises)
- **14:30-14:45:** Break
- **14:45-15:30:** Testing and debugging

Evening Block (1 hour):

- **19:00-20:00:** Review day's work, make notes, plan tomorrow
- Connect concepts across topics
- Review previous week's material for retention
- **NEW:** 10 minutes of mental math drills (once you reach Week 39)

9.2 Daily Structure for Part-Time Study (10-15 hrs/week)

Weekday Evenings (2 hours, 3-4 days/week):

- **19:00-20:00:** Theory study (1 section of notes)
- **20:00-21:00:** Practice problems (10-15 problems)
- **NEW:** 5 minutes mental math drills at end of session (Week 39+)

Weekend Days (4-6 hours each day):

- **Morning:** Complete one resource section with deep focus
- **Afternoon:** Implementation work or projects
- **Evening:** Review and consolidation + 15 min mental math (Week 39+)

9.3 Weekly Milestones & Checkpoints

Every Sunday Evening:

1. Review what you learned this week
2. Test yourself on key concepts without notes
3. Solve 5-10 review problems from previous weeks
4. Update your progress tracker
5. Plan next week's schedule and goals
6. **NEW:** Do 20 rapid-fire mental math drills (5 minutes, timed)

Every 4 Weeks (Monthly Review):

1. Comprehensive review of the month's content
2. Attempt 20-30 problems covering all topics from the month
3. Update your GitHub portfolio with completed projects
4. Reflect on what worked well and what needs adjustment
5. Adjust your study approach if falling behind
6. **NEW:** Complete one full mental math module (50 drills) to track speed improvement

9.4 Retention Strategies

1. **Spaced Repetition:** Review material after 1 day, 1 week, 1 month
2. **Active Recall:** Test yourself before reviewing notes
3. **Teach Others:** Explain concepts to friends or write blog posts
4. **Connect Concepts:** Explicitly link new material to previous knowledge
5. **Practice Variety:** Mix problem types rather than blocking by topic
6. **NEW:** Mental math daily - even 5 minutes maintains speed and confidence

10 Additional Resources & Tips

10.1 Recommended Books (Optional Supplements)

Mathematics:

- Shreve - "Stochastic Calculus for Finance" (Vol I & II)
- Joshi - "The Concepts and Practice of Mathematical Finance"

Derivatives:

- Hull - "Options, Futures, and Other Derivatives"
- Brigo & Mercurio - "Interest Rate Models"

Programming:

- Duffy - "Financial Instrument Pricing Using C++"

P&L & Risk:

- Rebonato - "Volatility and Correlation" (advanced)
- Gregory - "The xVA Challenge" (counterparty risk)

10.2 Online Communities & Forums

-  Quantitative Finance Stack Exchange
-  Wilmott Forums
-  Reddit: r/quant, r/algotrading
-  LinkedIn groups for quantitative finance

10.3 Practice Platforms

-  LeetCode - for coding interview prep
-  HackerRank - quantitative finance challenges
-  QuantNet - forums and resources
-  Kaggle - ML competitions with financial data

10.4 Tracking Your Progress

Create a tracking system for:

- Resources completed (with dates)
- Problems solved by topic and difficulty
- Projects completed with GitHub links
- Weak areas needing review
- Interview applications and responses
- **NEW:** Mental math speed metrics (problems/minute, accuracy %)
- **NEW:** P&L diagnostic case studies completed

10.5 Common Pitfalls to Avoid

Avoid These Common Mistakes:

1. **Rushing through theory:** Don't skip derivations to get to implementation
2. **Passive reading:** Always work through examples yourself
3. **Ignoring code quality:** Write production-level code from day one
4. **Studying in isolation:** Join study groups or find study partners
5. **Perfectionism:** Move forward even if you don't understand everything
6. **Neglecting review:** Regular review prevents forgetting
7. **Skipping projects:** Coding skills are as important as theory
8. **Not tracking progress:** You need to see your improvement
9. **NEW:** Skipping mental math practice because "you know the math"
10. **NEW:** Ignoring P&L attribution - it's what separates desk quants from academics

11 Final Checklist - Are You Ready?

Before starting your job search and interviews, ensure you can confidently answer YES to these questions:

11.1 Mathematical Foundations

- ☐ Can derive Itô's lemma and apply it to stochastic processes
- ☐ Understand martingales and why they matter for risk-neutral pricing
- ☐ Can solve the Black-Scholes PDE using multiple methods
- ☐ Comfortable with change of measure (Girsanov theorem)
- ☐ Can perform PCA on a yield curve and interpret results

11.2 Desk Skills & P&L Mastery

- ☐ Can separate clean vs dirty P&L and explain what each reveals
- ☐ Can diagnose whether P&L loss is execution, liquidity, or model slippage
- ☐ Understand when Greeks lie (barriers, high vol-of-vol, illiquid markets)
- ☐ Can detect correlation concentration using eigenvalue analysis
- ☐ Know that residual P&L $\geq 5\%$ means model failure, not P&L problem
- ☐ Can run 5-minute P&L triage: execution $\rightarrow$ dirty $\rightarrow$ Greeks $\rightarrow$ drift $\rightarrow$ new factor

11.3 Mental Math & Market Intuition

- ☐ Can bound any option in 10 seconds: $0 \leq C \leq S, 0 \leq P \leq K$
- ☐ Can identify vol regime (normal vs crisis) in 5 seconds from quotes
- ☐ Can use 0.4 rule to price ATM options within 10% in 5 seconds
- ☐ Can calculate FX crosses and detect triangular arbitrage in 15 seconds
- ☐ Never freeze - always use "Let me bound this first" protocol
- ☐ Can apply WRAPS framework to any trade quality question in 30 seconds

11.4 Asset Class Models

- ☐ Can calibrate Hull-White model to market instruments
- ☐ Understand when to use local vol vs stochastic vol
- ☐ Can explain FX smile dynamics and quanto adjustments
- ☐ Understand credit correlation and copula models
- ☐ Can articulate model limitations for each asset class

11.5 Products & Implementation

- ☐ Know payoff structures for 40+ derivative products
- ☐ Can price and hedge vanilla and exotic options
- ☐ Understand Greeks and their practical applications
- ☐ Have implemented 30+ projects with clean code
- ☐ Can explain P&L attribution for complex portfolios
- ☐ Every project outputs clean vs dirty P&L separation
- ☐ Can demonstrate residual P&L pattern detection in code

11.6 Interview Readiness

- ☐ Can solve 80%+ of interview problems under time pressure
- ☐ Have a GitHub portfolio with documented projects
- ☐ Can code pricers and risk metrics cleanly on whiteboard
- ☐ Prepared stories for behavioral questions
- ☐ Researched target firms and their trading strategies
- ☐ Can demonstrate mental math speed (30 sec per bound check)
- ☐ Have memorized vol ranges, $\sqrt{T}$ table, and power phrases

12 Resource Summary & Purchase Guide

12.1 Complete Resource List (26 Resources)

Phase 1: Foundation (3 Resources)

1. Quant ROADMAP (CAREER10)
2. Quant Finance Starter Pack (QUANT10)
3. Probability Quick Tricks (PROB10)

Phase 2: Mathematics + Python + Regulatory (6 Resources)

1. Linear Algebra & Differential Equations (LADE10)
2. Statistics & Econometrics (STATS10)
3. Stochastic Calculus (STOCHASTIC10)
4. Machine Learning for Quants (ML10)
5. **NEW:** Python for Quants (PYTHON10)
6. **NEW:** Regulatory & Risk Framework Primer (RISK10)

Phase 3: Desk Essentials (2 Resources)

1. Desk Essentials (DESK10)
2. **NEW:** P&L Attribution & Desk Diagnostics (PNL10)

Phase 4: Asset Class Models (4 Resources)

1. Interest Rates Models (RATES10)
2. FX Models (FXD10)
3. Equity Models (EQUITIES10)
4. Credit Models (CREDITS10)

Phase 5: Derivatives Products (5 Resources)

1. Interest Rate Derivatives - 15 products (IR10)
2. FX Derivatives - 12 products (FX10)
3. Equity Derivatives - 12 products (EQUITY10)
4. Credit Derivatives - 10 products (CREDIT10)
5. Inflation Derivatives - 8 products (INFLATION10)

Phase 6: Implementation (4 Resources)

1. Quant Projects Pack Part 1 - Foundational (PROJECT10)
2. Quant Projects Pack Part 2 - Moderate
3. Quant Projects Pack Part 3 - Advanced
4. Quant Projects Pack Part 4 - Asset-Specific

Phase 7: Interview Mastery (2 Resources)

1. **NEW:** Mental Math & Market Intuition Drills (MENTAL10)
2. 250+ Quant Interview Problems (FIRST10)

12.2 Strategic Purchase Timeline

Month 1 (Weeks 1-4): Buy: Roadmap, Starter Pack, Probability, LADE, Stats (5 resources) • Investment: \$150-200

Month 2 (Weeks 5-8): Buy: Stochastic, ML (2 resources) • Investment: \$60-80

Month 3 (Weeks 9-13): Buy: Desk Essentials, P&L Attribution (2 resources) • Investment: \$60-80

Month 4 (Weeks 14-17): Buy: IR Models, FX Models (2 resources) • Investment: \$60-80

Month 5 (Weeks 18-22): Buy: Equity Models, Credit Models (2 resources) • Investment: \$60-80

Month 6-7 (Weeks 23-30): Buy: All 5 derivative resources (5 resources) • Investment: \$150-200

Month 8 (Weeks 31-34): Buy: Projects Pack (1 resource) • Investment: \$30-40

Month 9-10 (Weeks 35-42): Buy: Mental Math, Interview Problems (2 resources) • Investment: \$60-80

Total Investment: \$650-850 with all coupon codes applied

13 Contact & Support

 **Questions about the materials?** Each resource includes contact information for clarifications and updates.

 **Direct Contact:** For personalized guidance, reach out via email at jha.8@alumni.iitj.ac.in

 **Follow on LinkedIn:** Stay updated with new resources and industry insights at [LinkedIn Profile](#)

 **Updates:** All resources are updated regularly with new content and corrections. Lifetime access includes all future updates.

 **Feedback:** Your feedback helps improve these materials. Use the provided channels to suggest improvements or report errors.

 **Community:** Join the growing community of learners working through this path. Share your progress, ask questions, and help others.

14 Success Stories & What Comes Next

14.1 After Completing This Path

You Will Have:

- Master's-level understanding of quantitative finance
- Portfolio of 40+ production-quality projects
- Ability to diagnose P&L like a senior model engineer
- Mental math speed that impresses in interviews
- Solutions to 250+ real interview questions
- Deep expertise in pricing, risk, and model implementation
- Confidence to interview at top-tier firms

Target Roles:

- Quantitative Analyst (Pricing, Risk, Structuring)
- Quantitative Trader (Systematic, Market Making)
- Quantitative Researcher (Alpha, Strategies)
- Model Validation Analyst
- Risk Analyst (Market Risk, XVA)
- Quantitative Developer (Trading Systems, Pricing Libraries)

Next Steps:

1. Polish your GitHub portfolio (add READMEs, documentation)
2. Create a professional website showcasing projects
3. Network on LinkedIn (connect with quants at target firms)
4. Apply to 20-30 positions aligned with your asset class focus
5. Continue learning - the field evolves rapidly
6. Give back - help others starting their journey

Deepen Expertise:

- Specialize in one asset class or modeling technique
- Contribute to open-source quant libraries
- Publish research or blog posts
- Mentor junior quants entering the field
- Build more advanced projects (ML-based pricers, full trading systems)

Good Luck on Your Quantitative Finance Journey!

Remember three principles:

1. Understanding WHY models fail is more valuable than knowing HOW they work
2. Residual P&L is the only honest feedback - everything else is self-consistency
3. In interviews, frameworks beat formulas - bound first, compute later

Stay curious, practice consistently, and build a portfolio that demonstrates your skills.